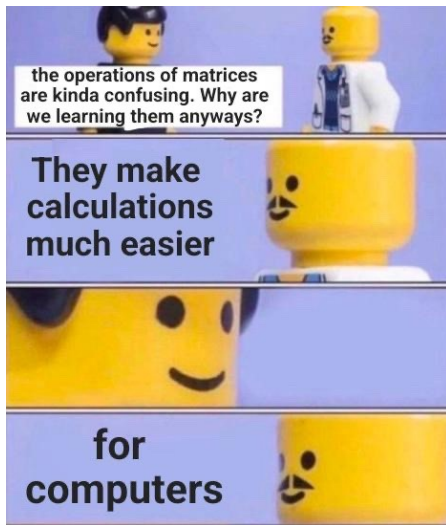


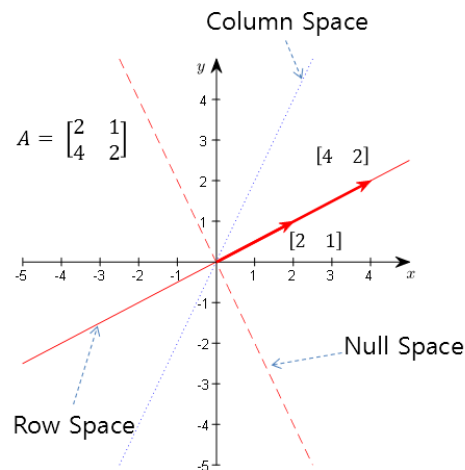
Solutions and Notes of Homework 3

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(a) A funny meme :)



(b) Can you see fundamental theorem of linear algebra (part 1 and part 2) here?

1. Let $M_{n \times n}$ be the vector space of all $n \times n$ matrices, and define $T : M_{n \times n} \rightarrow M_{n \times n}$ by $T(A) = A + A^T$.

(a) Show that T is a linear transformation.

Solution: For $A_1, A_2 \in M_{n \times n}$, we have $T(A_1 + A_2) = (A_1 + A_2) + (A_1 + A_2)^T = A_1 + A_1^T + A_2 + A_2^T = T(A_1) + T(A_2)$;

For $A \in M_{n \times n}$ and $c \in \mathbb{F}$, we have $T(cA) = (cA) + (cA)^T = c(A + A^T) = cT(A)$;

Hence T is a linear transformation.

(b) If $n = 2$:

(i) Show that the range of T is the set of B in $M_{2 \times 2}$ with the property that $B^T = B$.

Solution: Denote the set of B in $M_{2 \times 2}$ with the property that $B^T = B$ as U . It's easy to see that it is a subspace of $M_{2 \times 2}$. For any $C \in \text{range } T$, there exists $A \in M_{2 \times 2}$ such that $C = T(A) = A + A^T$. Then $C^T = (A + A^T)^T = A^T + A = C$, which implies that $C \in U$. Hence $\text{range } T \subset U$.

For any $B \in U$, $B = \frac{1}{2}B + \frac{1}{2}B^T = T\left(\frac{1}{2}B\right)$, which implies that $B \in \text{range } T$. Hence $U \subset \text{range } T$. This completes the proof.

(ii) Describe the null space of T .

Solution: Let $T(A) = A + A^T = 0$, then $A^T = -A$. Conversely, every skew-symmetric matrix B satisfies $T(B) = 0$. Hence $\text{null } T = \{A : A \in M_{2 \times 2}, A^T = -A\}$.

(iii) Find a basis of $\text{range } T$ and $\text{null } T$ and state the dimension of them.

Solution: Any matrix in $\text{range } T$ can be represented as $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$, which can further be represented as

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

which gives a spanning list of $\text{range } T$:

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Let

$$a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \implies a = b = c = 0$$

which implies the list is linearly independent. Hence it's a basis of $\text{range } T$, $\dim \text{range } T = 3$.

For $A \in \text{null } T$, suppose that $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

$$\begin{cases} a = -a \\ b = -c \\ c = -b \\ d = -d \end{cases} \implies a = d = 0, b = -c$$

Hence

$$A = \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix} = b \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

A single nonzero matrix is linearly independent, a basis of $\text{null } T$ is

$$\left\{ \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\},$$

$\dim \text{null } T = 1$.

(c) Solve (b)(iii) again when n is an arbitrary positive integer ($n \geq 2$).

Solution: Follow the steps in (b)(i) and (b)(ii) (which is not related to the size of matrices), it's easy to find that $\text{range } T$ is the subspace consisting of symmetric matrices and $\text{null } T$ is the subspace consisting of skew-symmetric matrices.

Any symmetric matrix can be represented as

$$\begin{aligned}
& \begin{bmatrix} a_1 & b_{1,2} & b_{1,3} & \cdots & b_{1,n-2} & b_{1,n-1} \\ b_{1,2} & a_2 & b_{2,3} & \cdots & b_{2,n-2} & b_{2,n-1} \\ b_{1,3} & b_{2,3} & a_3 & \cdots & b_{3,n-2} & b_{3,n-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ b_{1,n-2} & b_{2,n-2} & b_{3,n-2} & \cdots & a_{n-1} & b_{n-2,n-1} \\ b_{1,n-1} & b_{2,n-1} & b_{3,n-1} & \cdots & b_{n-2,n-1} & a_n \end{bmatrix} \\
&= a_1 \underbrace{\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n \text{ terms}} + \cdots + a_n \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}}_{n-1 \text{ terms}} \\
&+ b_{1,2} \underbrace{\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n-1 \text{ terms}} + \cdots + b_{1,n-1} \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n-2 \text{ terms}} \\
&+ b_{2,3} \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n-2 \text{ terms}} + \cdots + b_{2,n-1} \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n-2 \text{ terms}} \\
&+ \cdots \\
&+ b_{n-2,n-1} \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}}_{n-2 \text{ terms}}.
\end{aligned}$$

Putting the matrix after each coefficient together and we get a spanning set of range T . It's easy to verify the set is linearly independent. The dimension of range T is

$$n + (n-1) + \cdots + 1 = \frac{n(n+1)}{2}$$

Similarly, any skew-symmetric matrix can be represented as

$$\begin{aligned}
& \begin{bmatrix} 0 & b_{1,2} & b_{1,3} & \cdots & b_{1,n-2} & b_{1,n-1} \\ -b_{1,2} & 0 & b_{2,3} & \cdots & b_{2,n-2} & b_{2,n-1} \\ -b_{1,3} & -b_{2,3} & 0 & \cdots & b_{3,n-2} & b_{3,n-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -b_{1,n-2} & -b_{2,n-2} & -b_{3,n-2} & \cdots & 0 & b_{n-2,n-1} \\ -b_{1,n-1} & -b_{2,n-1} & -b_{3,n-1} & \cdots & -b_{n-2,n-1} & 0 \end{bmatrix} \\
&= b_{1,2} \underbrace{\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ -1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n-1 \text{ terms}} + \cdots + b_{1,n-1} \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n-2 \text{ terms}} \\
&+ b_{2,3} \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & -1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n-2 \text{ terms}} + \cdots + b_{2,n-1} \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 0 & \cdots & 0 & 0 \end{bmatrix}}_{n-3 \text{ terms}} \\
&+ \cdots \\
&+ b_{n-2,n-1} \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & -1 & 0 \end{bmatrix}
\end{aligned}$$

Putting the matrix after each coefficient together and we get a spanning set of $\text{null } T$. It's easy to verify the set is linearly independent. The dimension of $\text{null } T$ is $(n-1) + \cdots + 1 = \frac{n(n-1)}{2}$.

(d) 【选做】 Describe the relationship between $M_{n \times n}$, $\text{range } T$ and $\text{null } T$.

Solution: $M_{n \times n} = \text{range } T \oplus \text{null } T$. Verification:

(1) For any $A \in M_{n \times n}$, we have

$$A = \underbrace{\frac{A + A^T}{2}}_{\in \text{range } T} + \underbrace{\frac{A - A^T}{2}}_{\in \text{null } T}$$

which implies $\text{range } T + \text{null } T = M_{n \times n}$.

(2) Suppose that $A \in \text{range } T$ and $A \in \text{null } T$. Then $A = A^T$, $A = -A^T$, which implies $A = -A$. Therefore, $A = 0$, and thus $\text{range } T \cap \text{null } T = \{0\}$.

2. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_m$ is a list of vectors in V . Define $T \in \mathcal{L}(\mathbb{R}^m, V)$ by

$$T(\mathbf{x}) = x_1\mathbf{v}_1 + \cdots + x_m\mathbf{v}_m,$$

for $\mathbf{x} = \begin{bmatrix} x_1 & \cdots & x_m \end{bmatrix}^T \in \mathbb{R}^m$.

- (a) What property of T corresponds to $\mathbf{v}_1, \dots, \mathbf{v}_m$ spanning V ? (Injective or Surjective.) Why?
- (b) What property of T corresponds to $\mathbf{v}_1, \dots, \mathbf{v}_m$ being linearly independent? (Injective or Surjective.) Why?

Solution:

(a) Surjective.

If T is surjective, then $\text{range } T = V$, which implies for any $\mathbf{v} \in V$, there exists an $\mathbf{x} \in \mathbb{R}^m$ such that $T(\mathbf{x}) = \mathbf{v}$, thus every $\mathbf{v} \in V$ can be written as $\mathbf{v} = x_1\mathbf{v}_1 + \cdots + x_m\mathbf{v}_m$, i.e., a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_m$. Therefore, $\mathbf{v}_1, \dots, \mathbf{v}_m$ span V .

On the other hand, if $\mathbf{v}_1, \dots, \mathbf{v}_m$ span V , then for any $\mathbf{v} \in V$, we have $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m$ ($a_1, \dots, a_m \in \mathbb{R}$), which implies $T\left(\begin{bmatrix} a_1 & \cdots & a_m \end{bmatrix}^T\right) = \mathbf{v}$. So for any $\mathbf{v} \in V$, there exists a certain vector $\mathbf{x} \in \mathbb{R}^m$ such that $T(\mathbf{x}) = \mathbf{v}$, i.e., T is surjective.

(b) Injective.

If T is injective, then $\text{null } T = \{\mathbf{0}\}$, which implies $T(\mathbf{x}) = x_1\mathbf{v}_1 + \cdots + x_m\mathbf{v}_m = \mathbf{0}$ if and only if $\mathbf{x} = \mathbf{0}$, i.e., $x_1 = x_2 = \cdots = x_m = 0$. Therefore $\mathbf{v}_1, \dots, \mathbf{v}_m$ is linearly independent.

On the other hand, if $\mathbf{v}_1, \dots, \mathbf{v}_m$ is linearly independent, let $\mathbf{x} \in \text{null } T$, $T(\mathbf{x}) = x_1\mathbf{v}_1 + \cdots + x_m\mathbf{v}_m = \mathbf{0}$. Because $\mathbf{v}_1, \dots, \mathbf{v}_m$ is linearly independent, $x_1, \dots, x_m$ can only be 0, and $\mathbf{x}$ must be $\mathbf{0}$. Thus, $\text{null } T = \{\mathbf{0}\}$, which implies T is injective.

3. Let $E = \{u_1, u_2, u_3\}$ and $F = \{b_1, b_2\}$, where

$$u_1 = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad u_3 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

and

$$b_1 = (1, -1)^T, \quad b_2 = (2, -1)^T.$$

For each of the following linear transformations L from $\mathbb{R}^3$ into $\mathbb{R}^2$, find the matrix representing L with respect to the ordered bases E and F :

$$(i) L(x) = \begin{pmatrix} x_1 + x_2 \\ x_1 - x_3 \end{pmatrix}; \quad (ii) L(x) = \begin{pmatrix} 2x_2 \\ -x_1 \end{pmatrix}.$$

Solution:

(i)

$$L(u_1) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} = 3b_1 - b_2, \quad L(u_2) = \begin{pmatrix} 3 \\ 0 \end{pmatrix} = -3b_1 + 3b_2, \quad L(u_3) = \begin{pmatrix} 0 \\ -2 \end{pmatrix} = 4b_1 - 2b_2$$

Thus the matrix representation is $\begin{bmatrix} 3 & -3 & 4 \\ -1 & 3 & -2 \end{bmatrix}$.

(ii)

$$L(u_1) = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = 2b_1 - b_2, L(u_2) = \begin{pmatrix} 4 \\ -1 \end{pmatrix} = -2b_1 + 3b_2, L(u_3) = \begin{pmatrix} 2 \\ 1 \end{pmatrix} = -4b_1 + 3b_2$$

Thus the matrix representation is $\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 3 \end{bmatrix}$.

4. Find the standard matrices A of the following linear transformations.

- (a) $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ is a vertical shear transformation that leaves $\mathbf{e}_2$ unchanged and maps $\mathbf{e}_1$ into $2\mathbf{e}_2 + \mathbf{e}_1$.
- (b) $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ first performs a horizontal shear transformation that leaves $\mathbf{e}_1$ unchanged and maps $\mathbf{e}_2$ into $-2\mathbf{e}_1 + \mathbf{e}_2$ and then reflects points through the line $x_2 = x_1$.

Solution:

(a) $[\mathbf{e}_1 \quad \mathbf{e}_2] \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = [2\mathbf{e}_2 + \mathbf{e}_1 \quad \mathbf{e}_2]$, so the standard matrix is $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$.

(b) Denote the horizontal shear transformation as T_1 and the reflection as T_2 . $T_1(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $T_1(\mathbf{e}_2) = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$; $T(\mathbf{e}_1) = T_2(T_1(\mathbf{e}_1)) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $T(\mathbf{e}_2) = T_2(T_1(\mathbf{e}_2)) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. So the standard matrix is $\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$.

5. Let V be the subspace of $C[a, b]$ (continuous functions on $[a, b]$) spanned by $1, e^x, e^{-x}$, and let D be the differentiation operator on V .

- (a) Find the matrix A representing D with respect to the ordered basis $[1, \cosh x, \sinh x]$. [$\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$.]
- (b) Find the matrix B representing D with respect to $[1, e^x, e^{-x}]$.
- (c) Find the invertible matrix S such that $B = S^{-1}AS$. (Hint: transition matrix)

Solution:

(a)

$$\begin{aligned} D(1) &= 0 = 0 \cdot 1 + 0 \cdot \cosh x + 0 \cdot \sinh x \\ D(\cosh x) &= \frac{1}{2}(e^x - e^{-x}) = 0 \cdot 1 + 0 \cdot \cosh x + 1 \cdot \sinh x \implies A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \\ D(\sinh x) &= \frac{1}{2}(e^x + e^{-x}) = 0 \cdot 1 + 1 \cdot \cosh x + 0 \cdot \sinh x \end{aligned}$$

(b)

$$\begin{aligned} D(1) &= 0 = 0 \cdot 1 + 0 \cdot e^x + 0 \cdot e^{-x} \\ D(e^x) &= e^x = 0 \cdot 1 + 1 \cdot e^x + 0 \cdot e^{-x} \\ D(e^{-x}) &= -e^{-x} = 0 \cdot 1 + 0 \cdot e^x - 1 \cdot e^{-x} \end{aligned} \implies B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

(c) S is the transition matrix from $[1, e^x, e^{-x}]$ to $[1, \cosh x, \sinh x]$.

$$\begin{aligned} 1 &= 1 \cdot 1 + 0 \cdot \cosh x + 0 \cdot \sinh x \\ e^x &= 0 \cdot 1 + 1 \cdot \cosh x + 1 \cdot \sinh x \\ e^{-x} &= 0 \cdot 1 + 1 \cdot \cosh x - \sinh x \end{aligned} \implies S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

6. 【选做：几何拓展。不计入成绩】The *homogeneous* coordinate system is formed by equating each vector in $\mathbb{R}^2$ with a vector in $\mathbb{R}^3$ having the same first two coordinates and having 1 as its third coordinate.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

When we want to plot a point represented by the homogeneous coordinate vector $(x_1, x_2, 1)^T$, we simply ignore the third coordinate and plot the ordered pair (x_1, x_2) . Let

$$R = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

The column vectors of R represent the homogeneous coordinates of points in the plane.

(a) Draw the figure whose vertices correspond to the column vectors of R . What type of figure is it?

Solution: The original figure is a **unit square**, shown in Figure 2.

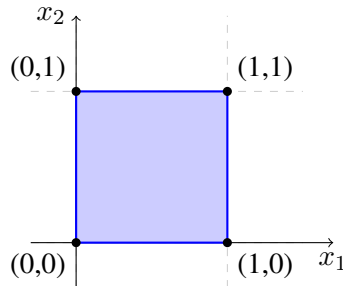
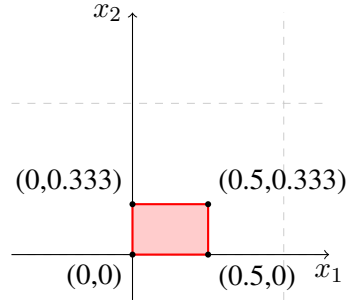


图 2: Original figure: unit square

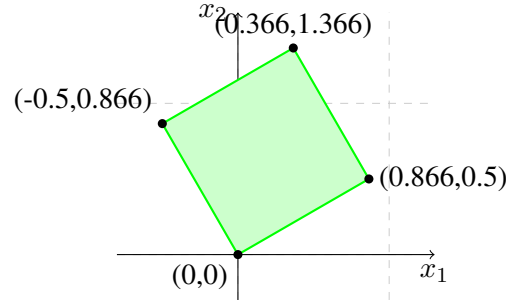
(b) For each of the following choices of A , sketch the graph of the figure represented by AR and observe the effect of the linear transformation. [Hint: pay special attention to (c). Think: why homogeneous coordinate?]

$$(1) A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad (2) A = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad (3) A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -2 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution: (1) This matrix scales the x_1 direction by $\frac{1}{2}$ and the x_2 direction by $\frac{1}{3}$. The transformed coordinates are $(0, 0)$, $(0, \frac{1}{3})$, $(\frac{1}{2}, \frac{1}{3})$, $(\frac{1}{2}, 0)$, $(0, 0)$, so the figure becomes a rectangle of width $1/2$ and height $1/3$ (Figure 3a).



(a) Anisotropic scaling: rectangle



(b) Rotation by 30° counterclockwise

图 3

(2) This is a counterclockwise rotation by 30° about the origin. The square is rotated but its shape and size remain unchanged (Figure 3b).

(3) The upper-left 2×2 block is an orthogonal matrix representing a clockwise rotation by 45° ; the first two entries of the last column, $(-2, 3)$, give a translation vector. The transformed points are

$$(-2, 3), \left(\frac{1}{\sqrt{2}} - 2, \frac{1}{\sqrt{2}} + 3 \right), (\sqrt{2} - 2, 3), \left(\frac{1}{\sqrt{2}} - 2, -\frac{1}{\sqrt{2}} + 3 \right), (-2, 3).$$

The figure is still a square, but now it has been rotated and translated (Figure 4).

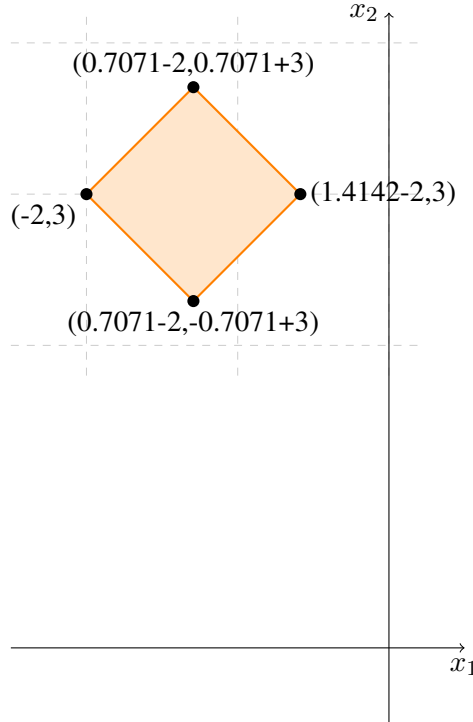


图 4: Rotation + translation (affine transformation)

Why homogeneous coordinates?

In ordinary 2D Cartesian coordinates, a translation cannot be written as a 2×2 matrix multiplication; it must be expressed as a vector addition $\mathbf{x}' = M\mathbf{x} + \mathbf{t}$. By introducing homogeneous coordinates, every 2D point (x_1, x_2) is extended to $(x_1, x_2, 1)^T$, and all affine transformations (rotation, scaling, translation, etc.) can be *unified* into a single 3×3 matrix multiplication:

$$\begin{bmatrix} \mathbf{x}' \\ 1 \end{bmatrix} = \begin{bmatrix} M & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}.$$

This uniform representation greatly simplifies the composition of transformations in computer graphics, which is exactly why homogeneous coordinates are fundamental. Part (c) above is a classic example: it combines rotation and translation in a single matrix product.

7. (a) Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \dots, v_n$ is linearly independent in V . Prove that $T(v_1), \dots, T(v_n)$ is linearly independent in W .
- (b) Suppose V is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace U of V such that $U \cap \text{null } T = \{0\}$ and $T(V) = T(U)$. Find a basis.

Solution:

(a) Since $v_1, \dots, v_n$ is linearly independent, then for scalars $a_1, \dots, a_n \in \mathbb{F}$,

$$a_1 v_1 + \dots + a_n v_n = 0 \iff a_1 = \dots = a_n = 0.$$

Because T is injective, only 0 can be mapped into 0. Then

$$T(a_1 v_1 + \dots + a_n v_n) = 0 \iff a_1 = \dots = a_n = 0.$$

From the linearity of T , we have $a_1 T(v_1) + \dots + a_n T(v_n) = T(a_1 v_1 + \dots + a_n v_n)$. Therefore $T(v_1), \dots, T(v_n)$ is linearly independent in W .

(b) Suppose that $\dim \text{null } T = m$, $\dim V = n$, $m \leq n$. Let $u_1, \dots, u_m$ be a basis of $\text{null } T$, which can be extended into a basis of V : $u_1, \dots, u_m, v_1, \dots, v_{n-m}$. For any $v \in V$, there exist scalars $a_1, \dots, a_m, b_1, \dots, b_{n-m} \in \mathbb{F}$ such that

$$\begin{aligned} T(v) &= T(a_1 u_1 + \dots + a_m u_m + b_1 v_1 + \dots + b_{n-m} v_{n-m}) \\ &= \underbrace{T(a_1 u_1 + \dots + a_m u_m)}_{=0} + T(b_1 v_1 + \dots + b_{n-m} v_{n-m}) \\ &= T(b_1 v_1 + \dots + b_{n-m} v_{n-m}) \end{aligned}$$

Let $U = \text{span}(v_1, \dots, v_{n-m})$, then $T(V) \subset T(U)$. Note that every $u \in U$ can be expressed as $b_1 v_1 + \dots + b_{n-m} v_{n-m}$ ($b_1, \dots, b_{n-m} \in \mathbb{F}$), so reverse the equality above and we show that $T(U) \subset T(V)$. It follows from Theorem 2.6.3 that $U \cap \text{null } T = \{0\}$. The basis of U consists of the extended vectors $v_1, \dots, v_{n-m}$. ■

8. (a) Suppose V is finite-dimensional, U is a subspace of V , and $S \in \mathcal{L}(U, V)$. Prove there exists an invertible operator $T \in \mathcal{L}(V)$ such that $T(u) = S(u)$ for every $u \in U$ if and only if S is injective.

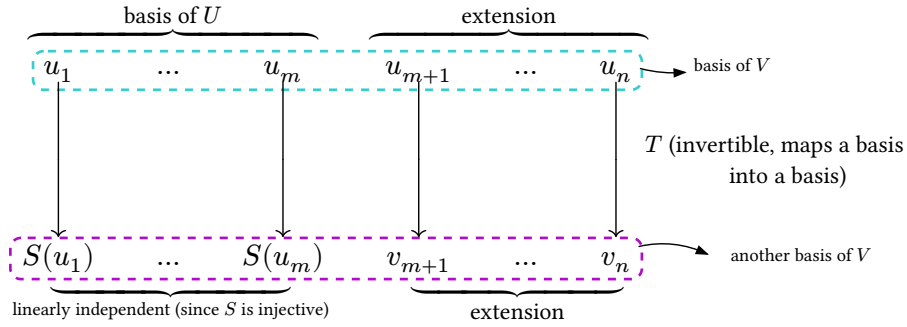
- (b) Suppose V, W are finite-dimensional and $T_1, T_2 \in \mathcal{L}(V, W)$. Prove that $\text{null } T_1 = \text{null } T_2$ if and only if there exists an invertible operator $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

Note: The key is to construct a linear operator/transformation through linear map(transformation) lemma.

Proof:

(a) First we assume there exists an invertible operator $T \in \mathcal{L}(V)$. Then T is injective. Then $T(u) = 0$ if and only if $u = 0$. Because $T(u) = S(u)$, we have $S(u) = 0$ if and only if $u = 0$. Thus $\text{null } S = \{0\}$, and S is injective, completing the first part of the proof.

Next we assume that S is injective.



The next corollary of Theorem 3.1.1 will be useful.

Corollary 0.1 Suppose that $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_n$ is a basis of W . Then the linear transformation $T : V \rightarrow W$ such that $Tv_i = w_i$ is invertible.

Let $u_1, \dots, u_m$ be a basis of U . Since U is a subspace of V , $u_1, \dots, u_m$ can be extended to a basis of V : $u_1, \dots, u_m, u_{m+1}, \dots, u_n$. Define $T \in \mathcal{L}(V)$ such that

$$T(u_1) = S(u_1), \dots, T(u_m) = S(u_m),$$

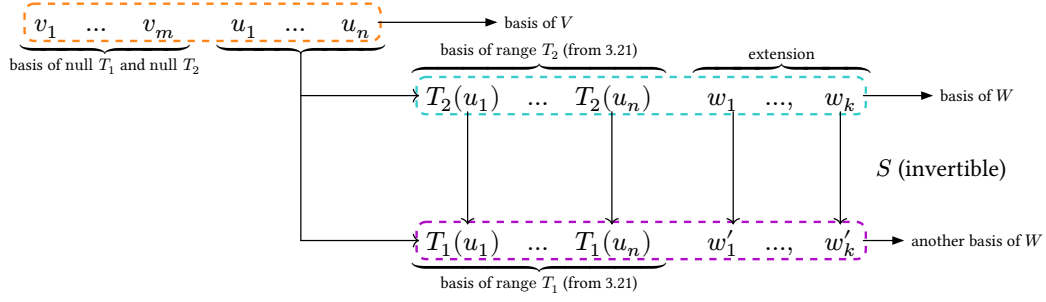
then $T(u) = S(u)$ for every $u \in U$. Since S is injective, $S(u_1), \dots, S(u_m)$ is linearly independent (refer to Problem 7(a)), so it can be extended to another basis of V : $S(u_1), \dots, S(u_m), v_{m+1}, \dots, v_n$. Let

$$T(u_{m+1}) = v_{m+1}, \dots, T(u_n) = v_n.$$

Thus T is well-defined. From corollary 0.1, T is invertible, completing the proof.

(b) First we assume that there exists an invertible operator $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$. Then for any vector $v \in \text{null } T_2$, $T_2(v) = 0$. Then $S(T_2(v)) = 0 = T_1(v)$, thus $\text{null } T_2 \subset \text{null } T_1$. For any vector $v \in \text{null } T_1$, $T_1(v) = S(T_2(v)) = 0$. Since S is injective, $T_2(v) = 0$. Thus $\text{null } T_1 \subset \text{null } T_2$. Hence $\text{null } T_1 = \text{null } T_2$, completing the first part of the proof.

Next we assume that $\text{null } T_1 = \text{null } T_2$.



Let $v_1, \dots, v_m$ be a basis of $\text{null } T_1$ and $\text{null } T_2$. Since $\text{null } T_1 (= \text{null } T_2)$ is a subspace of V , we can extend $v_1, \dots, v_m$ to a basis of V : $v_1, \dots, v_m, u_1, \dots, u_n$. T_1, T_2 can be written as follows:

$$T_1(v_1) = \dots = T_1(v_m) = 0, T_1(u_1), \dots, T_1(u_n),$$

$$T_2(v_1) = \dots = T_2(v_m) = 0, T_2(u_1), \dots, T_2(u_n),$$

where $T_1(u_1), \dots, T_1(u_n)$ and $T_2(u_1), \dots, T_2(u_n)$ are both lists of linearly independent vectors in W (from the proof of Theorem 3.4.1). Extend each list to a basis of W :

$$T_1(u_1), \dots, T_1(u_n), w_1, \dots, w_k \quad \text{and} \quad T_2(u_1), \dots, T_2(u_n), w'_1, \dots, w'_k,$$

and we can define $S \in \mathcal{L}(W)$ such that

$$S(T_2(u_1)) = T_1(u_1), \dots, S(T_2(u_n)) = T_1(u_n), S(w'_1) = w_1, \dots, S(w'_k) = w_k.$$

From corollary 0.1, S is invertible.

For any $v \in V$, there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n$ such that

$$\begin{aligned} ST_2(v) &= S(T_2(a_1v_1 + \dots + a_mv_m + b_1u_1 + \dots + b_nu_n)) \\ &= S(b_1T_2(u_1) + \dots + b_nT_2(u_n)) \\ &= b_1T_1(u_1) + \dots + b_nT_1(u_n) \\ &= a_1T_1(v_1) + \dots + a_mT_1(v_m) + b_1T_1(u_1) + \dots + b_nT_1(u_n) \\ &= T_1(v), \end{aligned}$$

hence $ST_2 = T_1$, completing the proof.

Note: To verify that ST_2 and T_1 are equal, you can also show that they map a basis into the same value.

9. Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $\dim \text{range } T = 1$ if and only if there exist a basis of V and a basis of W such that with respect to these bases, all entries of the matrix representation $\mathcal{M}(T)$ equal 1.

Proof: The key is to construct bases.

Suppose that $\dim V = n$ and $\dim W = m$.

$\Leftarrow$: Let $v_1, \dots, v_n$ be a basis of V and $w_1, \dots, w_m$ be a basis of W , which satisfy the condition.

Then we have

$$\begin{bmatrix} Tv_1 & Tv_2 & \cdots & Tv_n \end{bmatrix} = \begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix},$$

thus

$$Tv_1 = Tv_2 = \cdots = Tv_n = w_1 + \cdots + w_m.$$

Denote $w_1 + \cdots + w_m$ as w , thus $T(v) = T(k_1v_1 + \cdots + k_nv_n) = (k_1 + \cdots + k_n)w$. Hence $\text{range } T = \text{span}(w)$. Given that w is nonzero, $\dim \text{range } T = 1$, completing the first part of the proof.

$\implies$: From fundamental theorem of linear transformation (Theorem 3.4.1), $\dim V = \dim \text{null } T + \dim \text{range } T = \dim \text{null } T + 1$. Suppose that $v_1, \dots, v_{n-1}$ is a basis of $\text{null } T$. Extend it to a basis of V : $v_1, \dots, v_n$, and from Theorem 3.4.1 we know that $T(v_n)$ is a basis of $\text{range } T$. Denote it as w_1 . Extend the basis to form a basis in W : $w_1, \dots, w_m$. Then

$$\begin{bmatrix} Tv_1 & Tv_2 & \cdots & Tv_n \end{bmatrix} = \begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix} \begin{bmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & \cdots & 0 \end{bmatrix}_{m \times n}. \quad (1)$$

Proof 1: Choose a new basis of V :

$$v_1 + v_n, \dots, v_{n-1} + v_n, v_n, \quad (2)$$

so that every basis vector is mapped into the same value w_m . To verify that it is indeed a basis, let $a_1, \dots, a_n \in \mathbb{F}$, then for the equation

$$\begin{aligned} a_1(v_1 + v_n) + \cdots + a_{n-1}(v_{n-1} + v_n) + a_nv_n &= 0, \\ \implies a_1v_1 + \cdots + a_{n-1}v_{n-1} + \left(\sum_{i=1}^n a_i\right)v_n &= 0, \end{aligned}$$

and we can first find that $a_1 = 0$, and then $a_2 = 0, a_3 = 0, \dots, a_n = 0$. A list of linearly independent vectors that has appropriate length is a basis.

Then we choose a new basis of W :

$$w_1 - w_2 - \cdots - w_m, \dots, w_{m-1}, w_m, \quad (3)$$

so that $w_1 = (w_1 - w_2 - \cdots - w_m) + \sum_{i=2}^m w_i$, i.e., the sum of all the new basis vectors. To verify that it is indeed a basis, let $a_1, \dots, a_m \in \mathbb{F}$, then for the equation

$$\begin{aligned} a_1(w_1 - w_2 - \cdots - w_m) + a_2w_2 + \cdots + a_mw_m &= 0, \\ \implies a_1w_1 + (a_2 - a_1)w_2 + \cdots + (a_m - a_1)w_m &= 0, \end{aligned}$$

and we can first find that $a_1 = 0$, and then $a_2 = a_1 = 0, \dots, a_m = a_1 = 0$. A list of linearly independent vectors that has appropriate length is a basis.

Therefore, with (2) and (3),

$$T(v_1 + v_n) = \cdots = T(v_{n-1} + v_n) = T(v_n) = w_1 = (w_1 - w_2 - \cdots - w_m) + \sum_{i=2}^m w_i,$$

and the matrix representation with respect to these two bases is the matrix whose entries all equal 1. This completes the proof. $\blacksquare$

Proof 2: (The proof below shows the construction process using language of matrices. Compare it with the proof above.)

First we perform elementary column manipulation to (1) for a proper basis of V . Multiply
$$\begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 1 & \ddots & \\ 1 & 1 & \cdots & 1 \end{bmatrix}_{n \times n}$$
 on the right of both sides and the left hand side becomes

$$\begin{bmatrix} T(\sum_{i=1}^n v_i) & T(\sum_{i=2}^n v_i) & \cdots & T(v_n) \end{bmatrix}, \quad (4)$$

while the right hand side becomes

$$\begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}_{m \times n}.$$

Next, perform elementary row manipulation in the equation above to find a proper basis of W and to transform the matrix that have 1's in the top row into a matrix that has 1 everywhere:

$$\begin{aligned} & \begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}_{m \times n} \\ &= \begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix} \begin{bmatrix} 1 & & & \\ -1 & 1 & & \\ & & \ddots & \ddots \\ & & & -1 & 1 \\ & & & & -1 & 1 \end{bmatrix}_{m \times m} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}_{m \times m} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}_{m \times n} \\ &= \begin{bmatrix} w_1 - w_2 & w_2 - w_3 & \cdots & w_{m-1} - w_m & w_m \end{bmatrix} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}_{m \times n}. \end{aligned} \quad (5)$$

Finally, we verify that the new lists of vectors in the procedure above are basis of V and W respectively. First, consider the vectors $\sum_{i=1}^n v_i, \sum_{i=2}^n v_i, \dots, v_n$ is a basis of V . Let $a_1, \dots, a_n \in \mathbb{F}$, then for the

equation

$$a_1 \left(\sum_{i=1}^n v_i \right) + a_2 \left(\sum_{i=2}^n v_i \right) + \cdots + a_n v_n = 0,$$

$$\implies a_1 v_1 + (a_1 + a_2) v_2 + (a_1 + a_2 + a_3) v_3 + \cdots + \left(\sum_{i=1}^n a_i \right) v_n = 0,$$

and we can first find that $a_1 = 0$, and then $a_2 = 0, a_3 = 0, \dots, a_n = 0$. A list of linearly independent vectors that has appropriate length is a basis.

Then we verify that $w_1 - w_2, w_2 - w_3, \dots, w_{m-1} - w_m, w_m$ is a basis of W . Let $a_1, \dots, a_m \in \mathbb{F}$, then for the equation

$$a_1(w_1 - w_2) + a_2(w_2 - w_3) + \cdots + a_{m-1}(w_{m-1} - w_m) + a_m w_m = 0,$$

$$\implies a_1 w_1 + (a_2 - a_1) w_2 + (a_3 - a_2) w_3 + \cdots + (a_m - a_{m-1}) w_m = 0,$$

and thus $a_1 = 0, a_2 = a_1 = 0, \dots, a_m = a_{m-1} = 0$. A list of linearly independent vectors that has appropriate length is a basis.

Therefore, with (4) and (5), the matrix representation with respect to these two bases is the matrix whose entries all equal 1. This completes the proof. ■

Attention: You should always start at a random basis and construct the basis you need step-by-step. You can't just assume that $Tv_i = c_i w (c_i \neq 0)$, because you impose additional conditions to v_i .

Note: When I solved this problem for the first time (in spring semester of 2022), I didn't use the "extend" technique and just assumed that the basis of range T is w_1 and $Tv_1 = k_1 w_1, Tv_2 = k_2 w_1, \dots, Tv_n = k_n w_1$ and $k_1, \dots, k_s \neq 0, k_{s+1} = \cdots = k_n = 0$ ($s \leq n$, without loss of generality). Then

$$\begin{bmatrix} Tv_1 & Tv_2 & \cdots & Tv_n \end{bmatrix} = \begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix} \begin{bmatrix} k_1 & \cdots & k_s & 0 & \cdots & 0 \\ 0 & 0 & \cdots & \cdots & \cdots & 0 \\ \vdots & \vdots & & & & \vdots \\ 0 & 0 & \cdots & \cdots & \cdots & 0 \end{bmatrix}_{m \times n}.$$

I then multiplied $\begin{bmatrix} \frac{1}{k_1} & & & & & & \\ & \frac{1}{k_2} & & & & & \\ & & \ddots & & & & \\ & & & \frac{1}{k_s} & \frac{1}{k_s} & \cdots & \frac{1}{k_s} \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix}$ on the right side and get

$$\begin{bmatrix} T(\frac{1}{k_1} v_1) & T(\frac{1}{k_2} v_2) & \cdots & T(\frac{1}{k_s} v_s) & T(\frac{1}{k_s} v_s + v_{s+1}) & \cdots & T(\frac{1}{k_s} v_s + v_n) \end{bmatrix}$$

$$= \begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}_{m \times n}.$$

Compare this with the current proof, you can see that the “extend” technique provides clearer insight into the “structure” of the linear transformation.

10. Suppose that V is a vector space. Show that V and $\mathcal{L}(\mathbb{F}, V)$ are isomorphic vector spaces. (**Note:** V is not necessarily finite-dimensional, so you can’t utilize the dimension test, and you should try to construct an isomorphic linear map.)

Proof (1): Define a map $\Phi : \mathcal{L}(\mathbb{F}, V) \rightarrow V$ by

$$\Phi(T) = T(1)$$

where $T \in \mathcal{L}(\mathbb{F}, V)$. We will show that Φ is an invertible linear map.

Linearity. For any $T, S \in \mathcal{L}(\mathbb{F}, V)$ and $a \in \mathbb{F}$,

$$\Phi(T + aS) = (T + aS)(1) = T(1) + aS(1) = \Phi(T) + a\Phi(S).$$

Hence Φ is linear.

Injectivity. Suppose $\Phi(T) = 0$. Then $T(1) = 0$. Since every element of $\mathbb{F}$ is a scalar multiple of 1, for any $\alpha \in \mathbb{F}$ we have $T(\alpha) = \alpha T(1) = 0$. Thus T is the zero transformation, so $\text{null } \Phi = \{0\}$ and Φ is injective.

Surjectivity. Let $v \in V$. Define $T_v : \mathbb{F} \rightarrow V$ by $T_v(\alpha) = \alpha v$. Clearly T_v is linear because $T_v(\alpha + \beta) = (\alpha + \beta)v = \alpha v + \beta v = T_v(\alpha) + T_v(\beta)$ and $T_v(c\alpha) = c\alpha v = cT_v(\alpha)$. Moreover $\Phi(T_v) = T_v(1) = v$. Thus every $v \in V$ has a preimage under Φ , so Φ is surjective.

Therefore Φ is an invertible linear map, which establishes an isomorphism between $\mathcal{L}(\mathbb{F}, V)$ and V . ■

Proof (2): Define $\Phi : V \rightarrow \mathcal{L}(\mathbb{F}, V)$ as follows: for each $v \in V$, let $\Phi(v) = T_v$, where $T_v : \mathbb{F} \rightarrow V$ is given by $T_v(c) = cv$ for all $c \in \mathbb{F}$. Clearly T_v is linear, so $\Phi(v) \in \mathcal{L}(\mathbb{F}, V)$.

Linearity. For $v, w \in V$ and $\alpha \in \mathbb{F}$, and any $c \in \mathbb{F}$,

$$\Phi(v + \alpha w)(c) = T_{v+\alpha w}(c) = c(v + \alpha w) = cv + \alpha(cw) = T_v(c) + \alpha T_w(c) = (\Phi(v) + \alpha\Phi(w))(c).$$

Thus $\Phi(v + \alpha w) = \Phi(v) + \alpha\Phi(w)$, so Φ is linear.

Injectivity. Suppose $\Phi(v) = 0$ (the zero map). Then for every $c \in \mathbb{F}$, $cv = 0$. Taking $c = 1$ gives $v = 0$, hence $\text{null } \Phi = \{0\}$ and Φ is injective.

Surjectivity. Let $T \in \mathcal{L}(\mathbb{F}, V)$ be arbitrary. Set $v = T(1) \in V$. For any $c \in \mathbb{F}$,

$$T(c) = cT(1) = cv = T_v(c) = \Phi(v)(c).$$

Therefore $T = \Phi(v)$, proving Φ is surjective.

Since Φ is linear, injective, and surjective, it is an isomorphism. Hence $V \cong \mathcal{L}(\mathbb{F}, V)$. ■

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